

Matrix Embedding with LSBM and Hamming Codes

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Hamming(7,4) Example with LSBM

Embedding $k = 3$ message bits into $n = 7$ cover elements:

- **Message bits:** $\mathbf{m} = [1, 0, 1]^T$
- **Cover Block:** $\mathbf{x} = [155, 144, 133, 122, 111, 100, 99]^T$
- **Extracted LSBs:** $\mathbf{v} = \mathbf{x} \bmod 2 = [1, 0, 1, 0, 1, 0, 1]^T$
- **Parity Matrix $\mathbf{H}$:** (Columns represent binary 1 to 7)

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

- **Syndrome:** $\mathbf{s} = \mathbf{H} \cdot \mathbf{v} \bmod 2 = [0, 0, 0]^T$

Hamming(7,4) Example with LSBM (cont.)

Matrix Embedding Logic:

$$\mathbf{d} = \mathbf{m} - \mathbf{s} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

The difference vector $\mathbf{d}$ matches **Column 5** of $\mathbf{H}$.

Action: Modify the 5th cover element (111 $\rightarrow$ 110 or 112). The new LSB sequence will yield exactly $\mathbf{m}$ when multiplied by $\mathbf{H}$.

Calculating Embedding Efficiency for given α

Core Task: Calculate the embedding efficiency (e) for a given target capacity (α).

Problem: Hamming codes only provide discrete capacities ($\alpha_k = \frac{k}{2^k-1}$). For "impossible" target capacities like $\alpha = 0.50$, we must linearly interpolate between the two neighboring codes.

Calculating Embedding Efficiency: A Mathematical Example

Step 1: Find Bounding Block Sizes (k and $k + 1$)

- $k = 2$: $\alpha_2 = 2/3 \approx 0.667$, $\beta_2 = 1/4 = 0.250$
- $k = 3$: $\alpha_3 = 3/7 \approx 0.429$, $\beta_3 = 1/8 = 0.125$

Step 2: Calculate the Mixing Weight (w)

We calculate the fraction of the image embedded with $k = 2$ blocks:

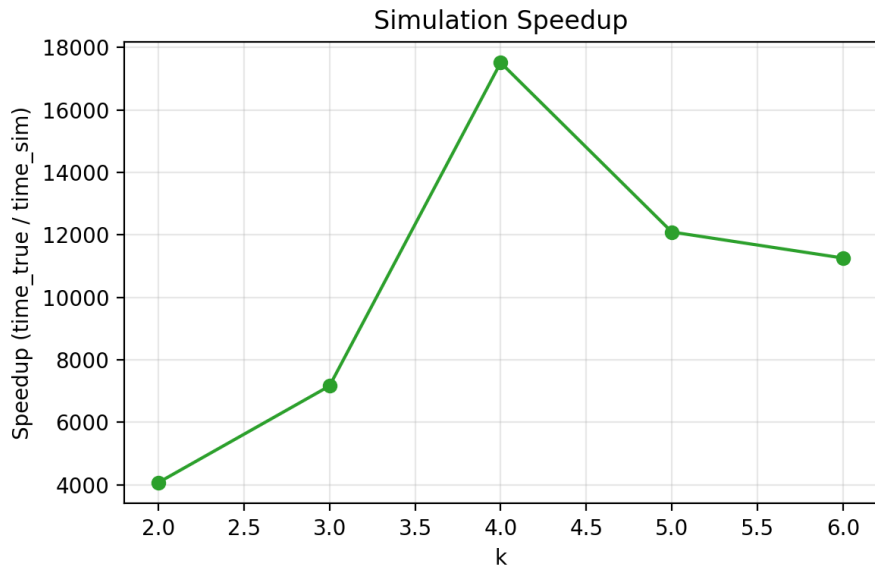
$$w = \frac{\alpha - \alpha_3}{\alpha_2 - \alpha_3} = \frac{0.50 - 0.429}{0.667 - 0.429} \approx 0.30 \quad (30\% \text{ of capacity from } k = 2)$$

Step 3: Calculate (β) and (e)

$$\beta_{\text{target}} = w\beta_2 + (1 - w)\beta_3 = 0.30(0.250) + 0.70(0.125) = 0.1625$$

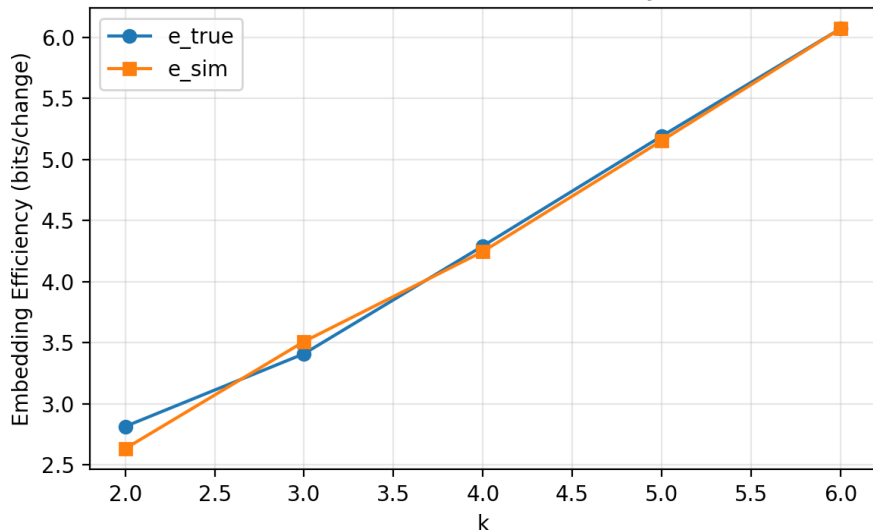
$$e = \frac{\alpha}{\beta_{\text{target}}} = \frac{0.50}{0.1625} \approx 3.08 \text{ bits/change}$$

Execution time for simulation and real embedding



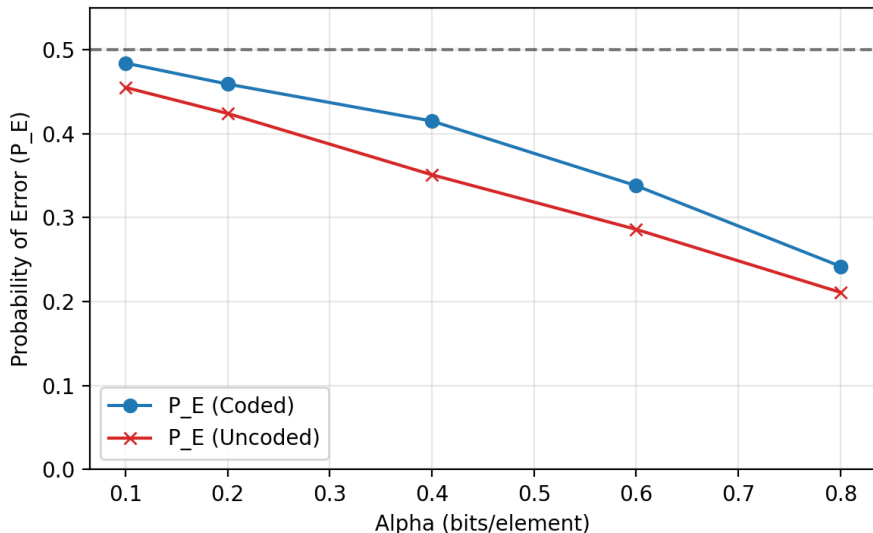
Embedding efficiency ε for different k

True vs Simulated Efficiency



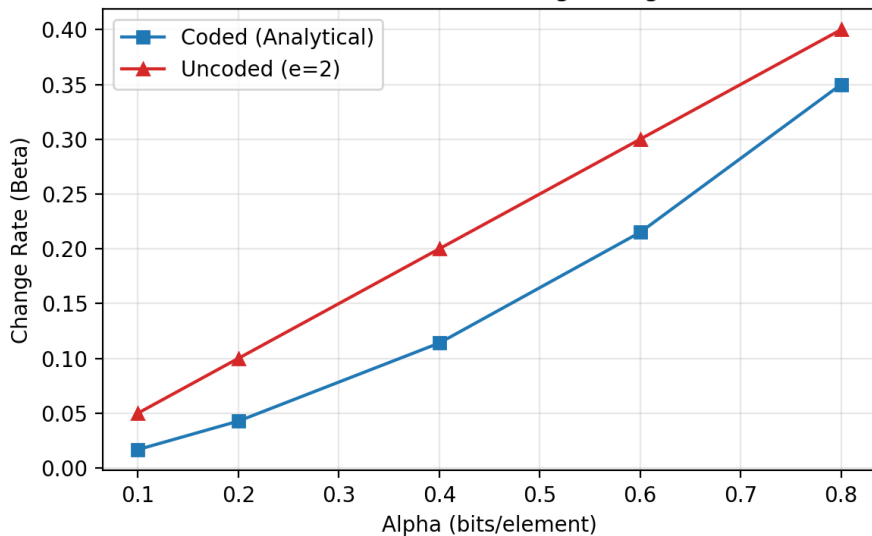
LSBM hamming against uncoded LSBM simulations

Simulated HCF-COM Attack Performance

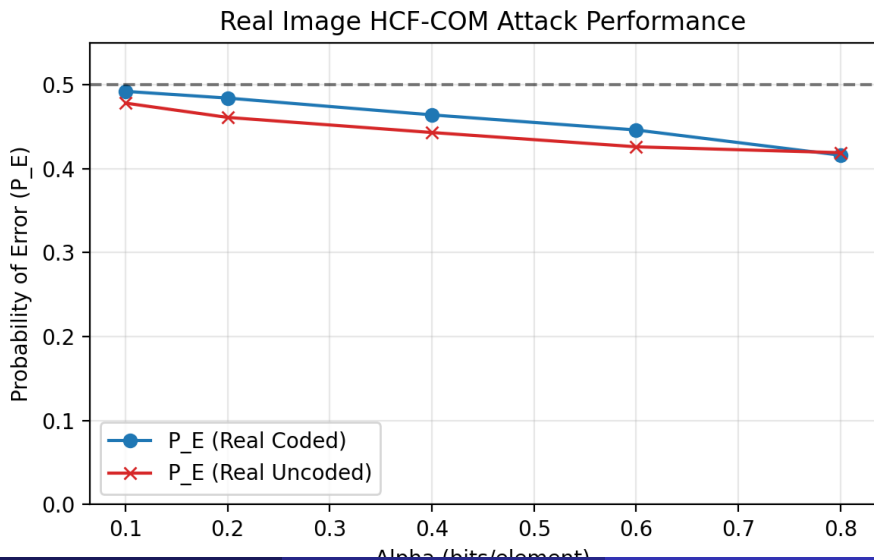


Change rate β for different k

Simulated Embedding Changes



LSBM hamming against uncoded LSBM 500 grayscale images



Summary

- **Completed Implementation:** Fully functional LSB Matching based upon matrix embedding with Hamming codes.
- **Fast Simulator derived:** Allows generating theoretical impacts for arbitrary embedding rates α via local interpolation.
- **Results:** Hamming codes reduce the change rate β therefore improve detectability against HCF.